

Geographic Dispersion of Economic Shocks: Evidence from the Fracking Revolution

Replication & Extension: Nonparametric Methods and Quantile Regression

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Nonparametric Methods and Distributional Analysis – June 2026

Based on: Feyrer, Mansur & Sacerdote, *AER* 2017

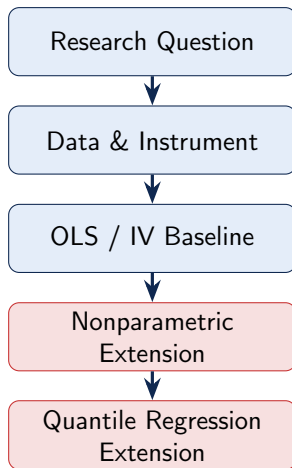
Presentation Roadmap

Original Paper

1. Setting & question
2. Data and instrument
3. Main OLS/IV results (Table 2)

Our Extensions

4. Distribution shifts (KDE + CDF)
5. Linearity tests (local polynomial)
6. Quantile regression: OLS-QR & IVQR
7. Quantile process across industries



The Original Paper: Overview

Question: How does a local energy boom propagate wages across industries and distances?

- **Setting:** US fracking revolution, 2004–2012
- **Outcome:** Δw_{ict} = annual change in wages per capita, 8 BLS industries
- **Treatment:** Value of oil & gas wells opened (\$M per capita)
L0 = current year; L1 = one-year lag

$$\Delta w_{ict} = \alpha_c + \delta_t + \beta_0 \text{NewVal}_{ct} + \beta_1 \text{NewVal}_{c,t-1} + \varepsilon_{ict}$$

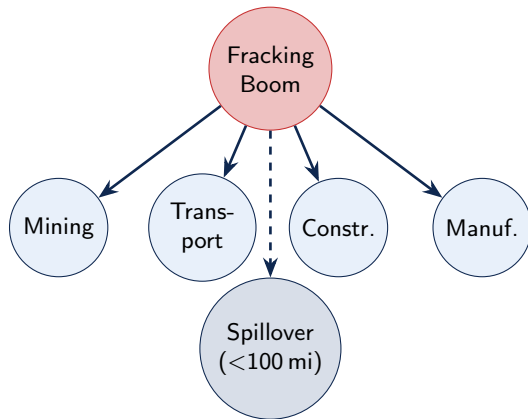
Δw_{ict} wages per capita, industry i , county c , year t

NewVal_{ct} new extraction value (\$M per capita)

α_c, δ_t county and year fixed effects

Key Finding (IV)

\$1M of new fracking output per county $\Rightarrow$ \$79,751 of county wages over 2 years;
spillovers decay steeply with distance.



Wage effects flow to multiple *non-extractive* sectors

Data Sources

- BLS QCEW (county $\times$ industry $\times$ year)
- IRS individual income files (wages per capita)
- County-level oil & gas production values

Instrument: Geology $\times$ national commodity price

- Local shale geology determines *potential* extraction
- Interacted with national prices — exogenous to local wages
- **Relevance:** strong first stage ($F \gg 10$)
- **Exclusion:** geology affects wages *only through* production

OLS likely attenuated: measurement error in production values

Sample

	Value	Note
Counties	$\approx 2,693$	Balanced panel
Years	8	2005–2012
Industries	8	BLS sectors
Obs. per ind.	21,546	County $\times$ year
Total obs.	172,368	$21,546 \times 8$

Our Results: L0+L1 Coefficient Sums by Industry & Quantile

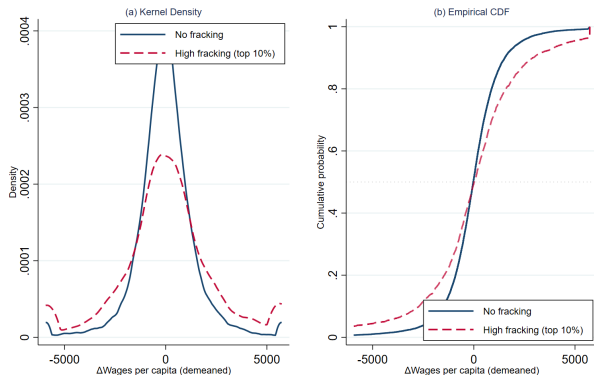
Δ Wages per capita. Reported: $\hat{\beta}_{L0} + \hat{\beta}_{L1}$ sum. Stars reflect significance of $\hat{\beta}_{L0}$ (dominant term). County + year FE (Canay 2011); bootstrap SEs.

Industry	Q0.10	Q0.50	Q0.90	OLS
Total	25,122***	27,131***	35,827***	38,674***
Mining	3,933**	9,923***	5,794***	16,034***
Transport	5,388***	4,736***	6,659***	10,122***
Construction	1,454	430**	3,233***	4,229***
Manufacturing	615	-2***	-1,499**	-776
Educ/Health	-204	90**	164	-972
Other Services	2,933***	3,394***	4,705***	4,151***
Government	5,552**	2,896***	4,731*	5,887

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Extension I — Distribution Shifts (KDE + CDF)

Wage Growth Distribution by Fracking Intensity — Total



Treatment = NewValue_t + NewValue_(t-1). County + year FE demeaned; wages winsorised 1/99 pct.
Source: Feyrer, Mansur & Sacerdote (2015).

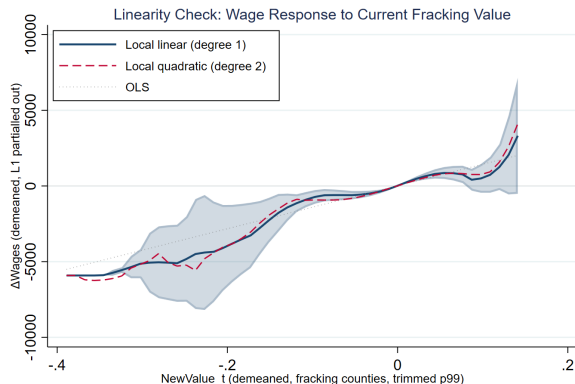
What we do:

- Classify counties into *No-fracking* vs. *High-fracking* (top 10% of L0+L1)
- Compare wage-growth distributions after two-way FE demeaning

Findings:

- High-fracking distribution is **more dispersed** — lower peak, heavier tails on both sides
- CDF lies to the right above the median — more probability mass at higher wage growth values
- **K-S test significant** ($p < 0.05$) — distributions differ primarily in **spread and upper tail**, not a simple mean shift

Extension II — Testing Linearity (Local Polynomial)



Total wage response. L1 partialled out (FWL).

Method: Local polynomial regression (Epanechnikov kernel; degrees 1 & 2 vs. OLS)

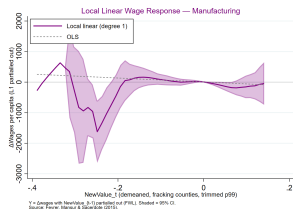
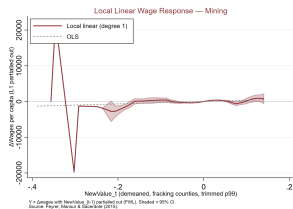
Key Observation

Local linear and quadratic fits **track each other closely**, and both are broadly consistent with the OLS slope.

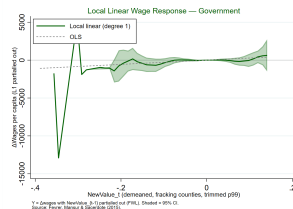
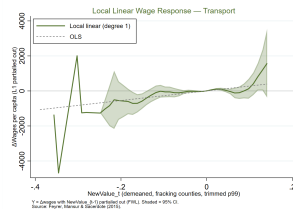
Implication:

- The original paper's log-linear specification *appears well-supported*
- No clear threshold or concavity in the data
- Some widening of CI at very high treatment values (sparse data)

Extension II — Local Linear by Industry



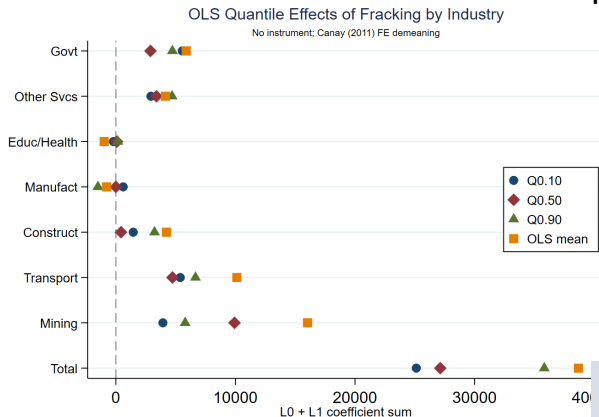
Mining & Transport: positive slopes, OLS close to local fit



Manufacturing: downward-sloping fit at high values: crowding-out signal

Extension III — Quantile Regression Coefficient Plots

OLS Quantile Regression (no IV)

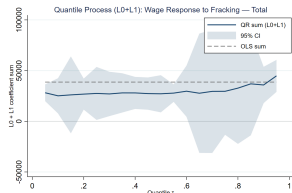


Each dot = L0+L1 at Q0.10/Q0.50/Q0.90 vs. OLS
Patterns:

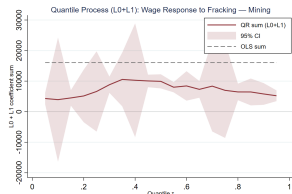
- **Directly exposed industries** (Mining, Transport): OLS mean lies *above* all three quantiles — driven by a small number of high-treatment counties
- **Mining**: non-monotone — rises Q10→Q50, falls at Q90
- **Manufacturing**: turns negative at Q50+ — Dutch Disease
- **Educ/Health & Govt**: small, mixed-sign — limited spillover to these sectors

Key Takeaway: OLS overstates the typical effect for exposed industries; heterogeneity is masked.

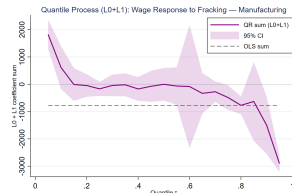
Extension IV — Quantile Process ($\tau \in [0.05, 0.95]$)



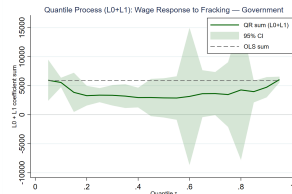
Sum of current + lagged coefficients across the conditional wage-growth distribution.
County + year FE demeaned, county-clustered SIs (pqr2). Source: Feyer et al. (2015).



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Dashed = OLS sum; shaded = 95% CI (county-clustered, qreg2).

Manufacturing: QR sum dips negative at lower quantiles — OLS misses this

Synthesis: Similarities, Differences, and Contribution

Where We Agree with FMS (2017)

- Strong positive mean effects in mining, transport, construction
- Conditional mean approximately linear — OLS functional form defensible *for mean regression*

Where OLS is Insufficient

- **K-S test** rejects distributional equality — fracking disperses the distribution
- **Quantile process** shows significant effect heterogeneity across τ that OLS averages away
- **Manufacturing:** OLS (-776 , insig.) masks a negative, significant effect across most of the distribution

Bottom Line

OLS gives an **incomplete and in Manufacturing a misleading** picture. Distributional heterogeneity is real, industry-specific, and invisible to mean regression.

Appendix: KDE by Industry

